



Mahidol University
Wisdom of the Land

Antifungal Drugs

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Pharmacology II (PYCH305)

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Learning Objectives

Students should be able to:

1. Explain the pharmacological properties of antifungals, including
 - Mechanism of action
 - Spectrum of activity
 - PK/PD considerations
 - Adverse drug reactions
 - Significant drug interactions
2. Select appropriate antifungals, based on their pharmacological properties, for the treatment of mycoses

Fungi

- Eukaryotes
 - Contain cell wall

Differ from humans
and animals

Dimorphic fungi

- *Blastomyces*
- *Coccidioides*
- *Histoplasma*
- *Sporothrix*

Molds

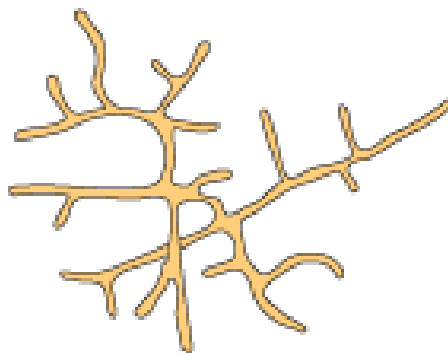
- *Aspergillus*
- *Fusarium*
- Zygomycetes
- Dermatophytes

Yeasts

- *Candida*
- *Cryptococcus*
- *Trichosporon*
- *Malassezia*

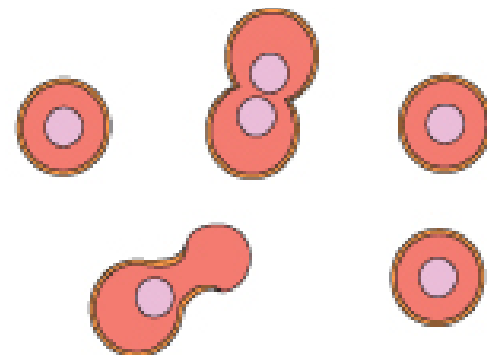
Filamentous

Growing as multinucleate,
branching hyphae, forming a
mycelium



Yeasts

Growing as ovoid or
spherical single cells,
multiply by budding and
division





Fungal infection (Mycotic infection; Mycosis)

Site of infection	Organ	Most frequent genus	Estimated incidence worldwide (cases/yr)
Superficial	Skin and hair	<i>Malassezia</i>	~140,000,000
Cutaneous	Skin, hair, and nail	<i>Trichophyton</i> <i>Epidermophyton</i> <i>Microsporum</i>	~1,500,000,000
Mucosal	Vagina	<i>Candida</i>	~75,000,000
	Digestive tract, urinary tract	<i>Candida</i>	~9,500,000
	Eye	<i>Candida, Aspergillus, Fusarium</i>	~1,000,000
Systemic	Internal organs including bloodstream	<i>Candida</i> <i>Aspergillus</i> <i>Cryptococcus</i> <i>Histoplasma</i> <i>Pneumocystis</i> <i>Coccidioidomyces</i>	~300,000 ~350,000 ~1,000,000 ~500,000 >200,000 Up to 300,000

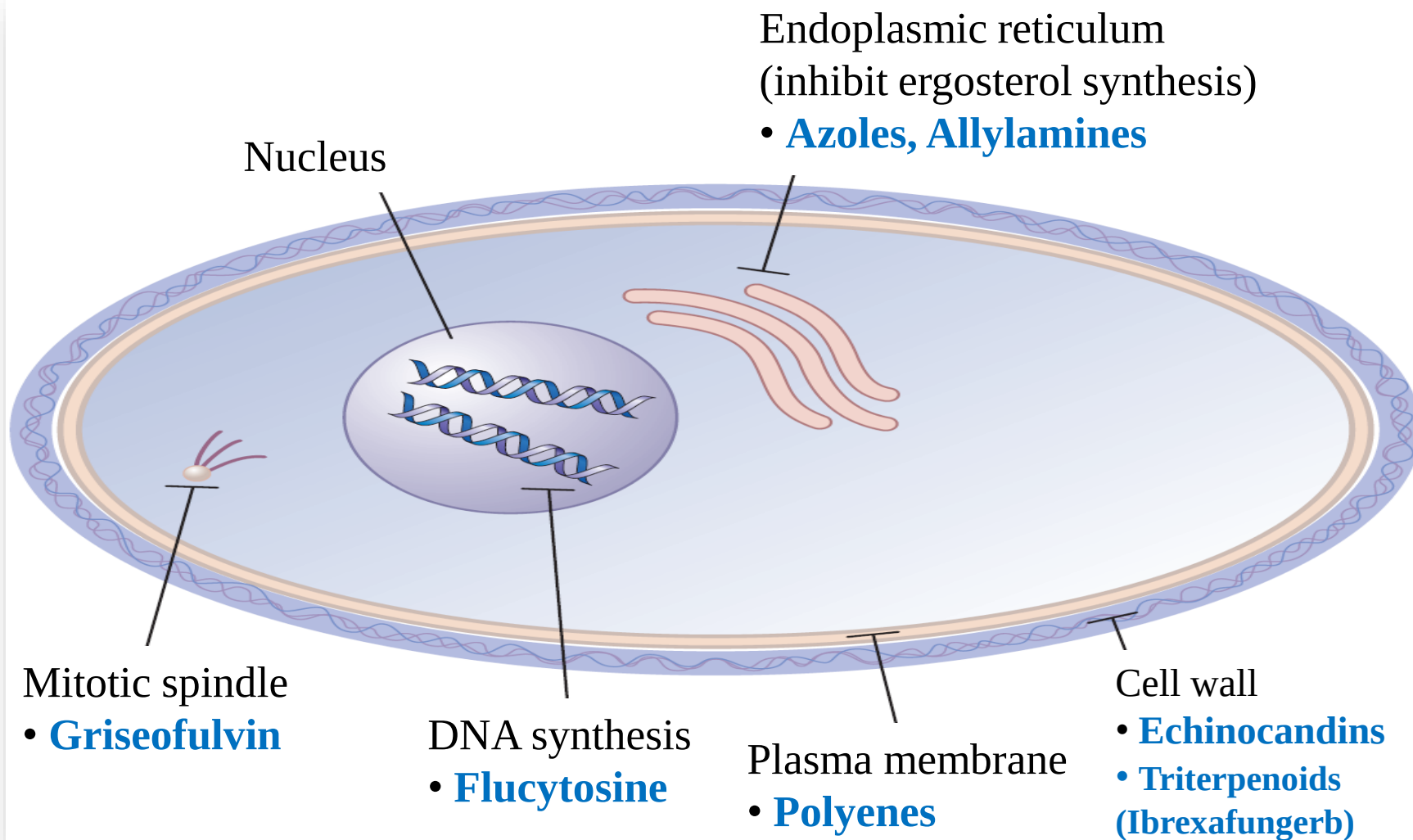


Clinical Uses of Antifungals

- Classified by Site of Infection

- Superficial mycoses
 - Cutaneous mycoses
- } Topical antifungals
Except: Severe cases,
Tinea unguium, Tinea capitis
- Subcutaneous mycoses
 - Systemic/Invasive/Deep mycoses
- } Systemic antifungals

Classification of Antifungals (Antimycotics)



Polyenes

- **Drugs:** Amphotericin B, Nystatin, Natamycin

- Mechanism of action

Bind “Ergosterol” of fungal cell membrane



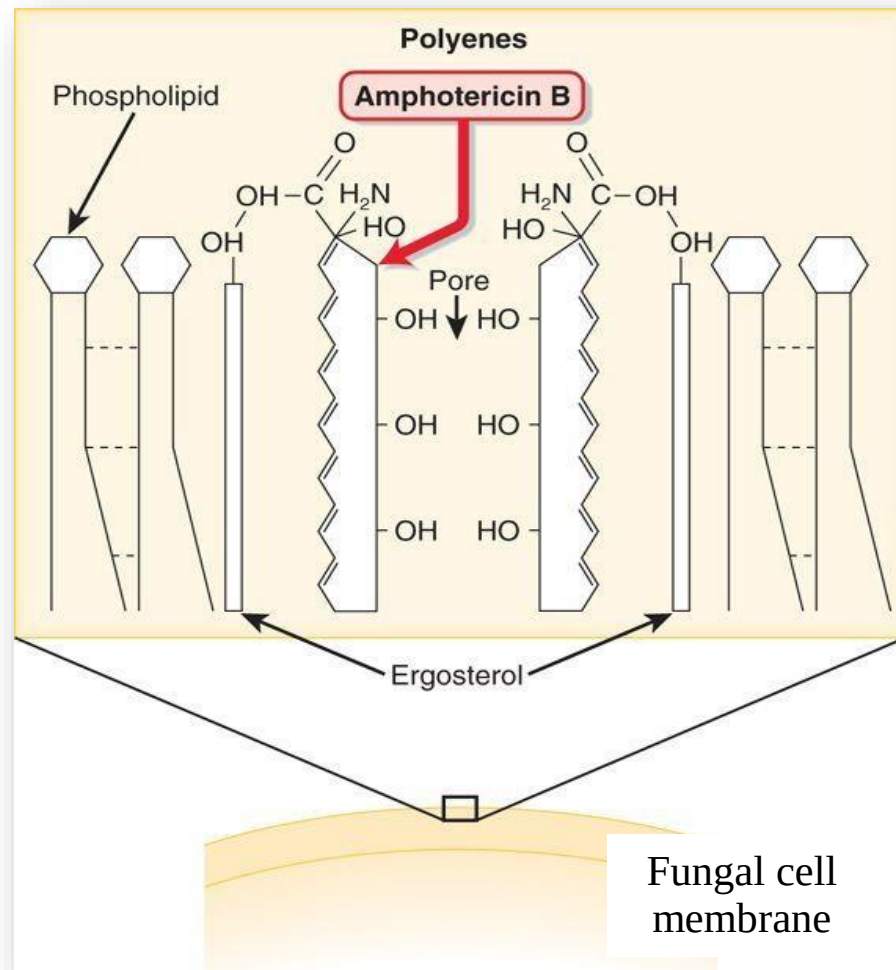
Pore forming



Small molecules leak out



Cell death





Amphotericin B

- Firstly isolated from *Streptomyces nodosus*
 - Poor GI absorption → Intravenous infusion
 - Long $t_{1/2}$ (> 50 hrs)
 - Wide distribution throughout the body
 - Low Conc: CSF, vitreous humor, amniotic fluid
 - “Conc. at brain parenchyma is believed to be sufficient for CNS mycoses”
 - Intrathecal, intraocular, intra-articular injection
 - Eye drop (freshly prepared), Bladder irrigation
- } In some cases



Amphotericin B: Spectrum of Activity

Organisms	Activity	Organisms	Activity
<i>Candida</i> spp.		<i>Aspergillus</i> spp.	
<i>C. albicans</i>	+	<i>A. flavus</i>	±
<i>C. glabrata</i>	+	<i>A. fumigatus</i>	+
<i>C. krusei</i>	+	<i>A. niger</i>	+
<i>C. lusitaniae</i>	-	<i>A. terreus</i>	-
<i>C. parapsilosis</i>	+	<i>A. nidulan</i>	+
<i>C. tropicalis</i>	+		
<i>C. guilliermondii</i>	-		

Note: + has activity, - does not have activity, ± variable activity



Amphotericin B: Spectrum of Activity (Cont.)

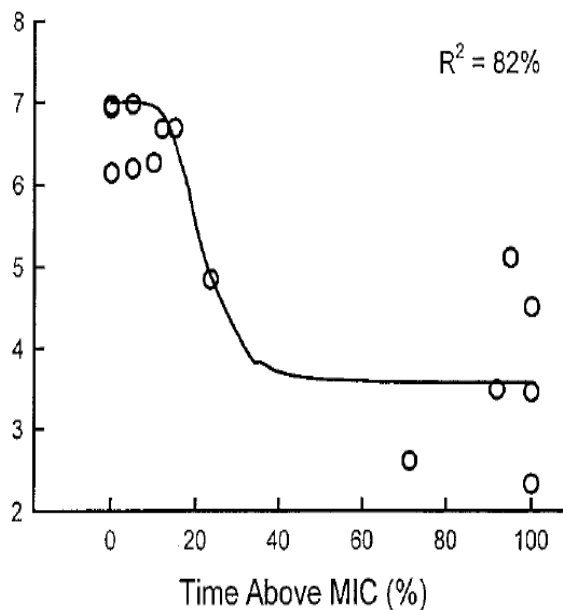
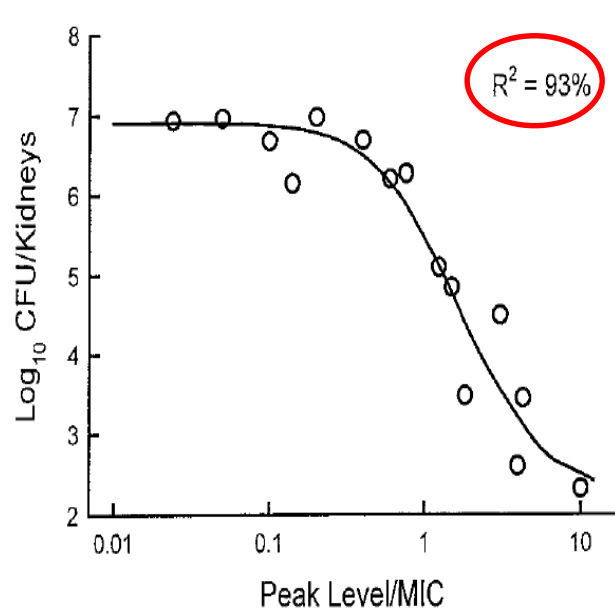
Organisms	Activity
<i>Cryptococcus neoformans</i>	+
<i>Coccidioides</i> spp.	+
<i>Blastomyces</i> (genus)	+
<i>Histoplasma</i> spp.	+
<i>Fusarium</i> spp.	±
<i>Scedosporium apiospermum</i> (asexual state) Or <i>Pseudallescheria boydii</i> (sexual state)	±
Zygomycetes (class), including Mucorales (order)	±

Note: + has activity, - does not have activity, ± variable activity



Amphotericin B: Pattern of Activity

- Fungicidal
- Conc.-dependent killing activity with prolonged PAE
 - Plateau at Conc. $> 4-10 \times \text{MIC}$



AmphoB deoxycholate:

- $\text{C}_{\text{max}}:\text{MIC} > 4-10$
Or
- $\text{AUC}:\text{MIC} > 100$

Liposomal AmphoB:

- $\text{C}_{\text{max}}:\text{MIC} > 40$
Or
- $\text{AUC}:\text{MIC} > 100$

Amphotericin B: Adverse Drug Reactions

Infusion-related reactions (70-90%)

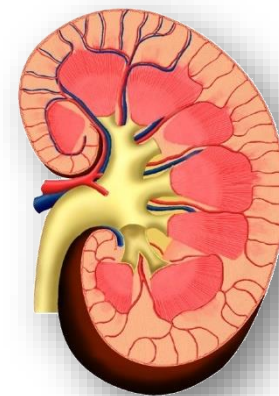
- **Fever, chills**, rigors, vomiting, hypo/hypertension
 - Proinflammatory cytokines release
 - Spontaneously subsides in 30-45 min
- Prevention
 - **Premedication**: 30 min before AmphoB
Acetaminophen + antihistamines Or IV hydrocortisone
 - **Infusion time** > 6 hrs is recommended
- Meperidine (Pethidine) may shorten the reaction, esp. rigors



Amphotericin B: Adverse Drug Reactions

Nephrotoxicity (24-80%)

- **Dose-dependent** (usually reversible)
- Azotemia, polyuria, acidosis
- **Mechanism of toxicity**
 - Vasoconstriction of afferent arteriole ➡ **Renal hypoperfusion**
 - May directly toxic to aff. arteriole and renal tubules
- Hydration with 1 L of NSS infusion (e.g. 500 ml before and after AmphoB) ➡ **Reduces incidence**
- Monitor kidney function (serum creatinine)

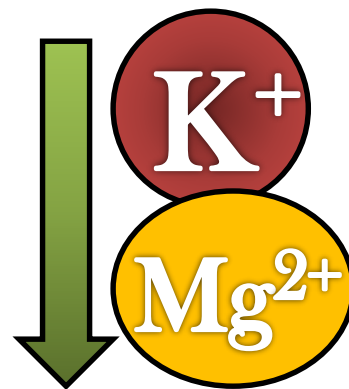


Bind
Cholesterol

Amphotericin B: Adverse Drug Reactions

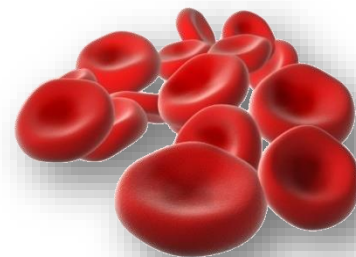
Hypokalemia (25-75%)
Hypomagnesemia (30-75%)

- Related with nephrotoxicity
Weakness → Arrhythmia
- Monitor serum K^+ and Mg^{2+} level
- K^+ supplement may be required if prolonged AmphoB uses



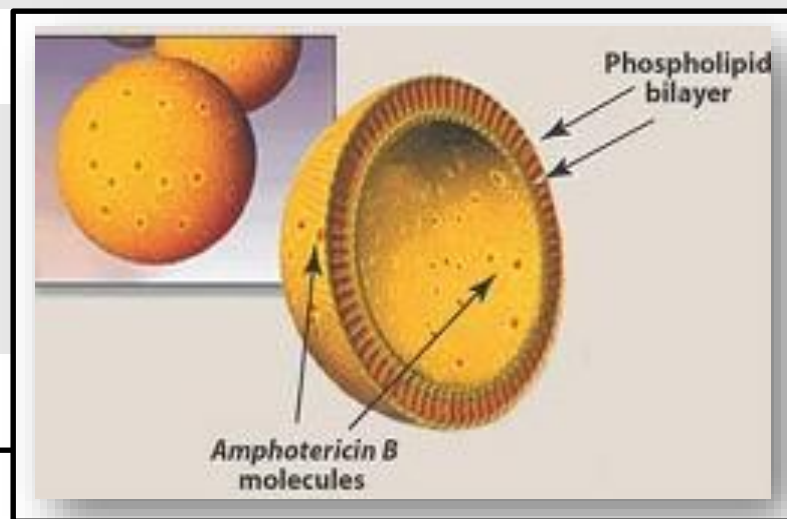
Anemia (~75%)

- Due to ↓ EPO
- Hypochromic, normocytic anemia



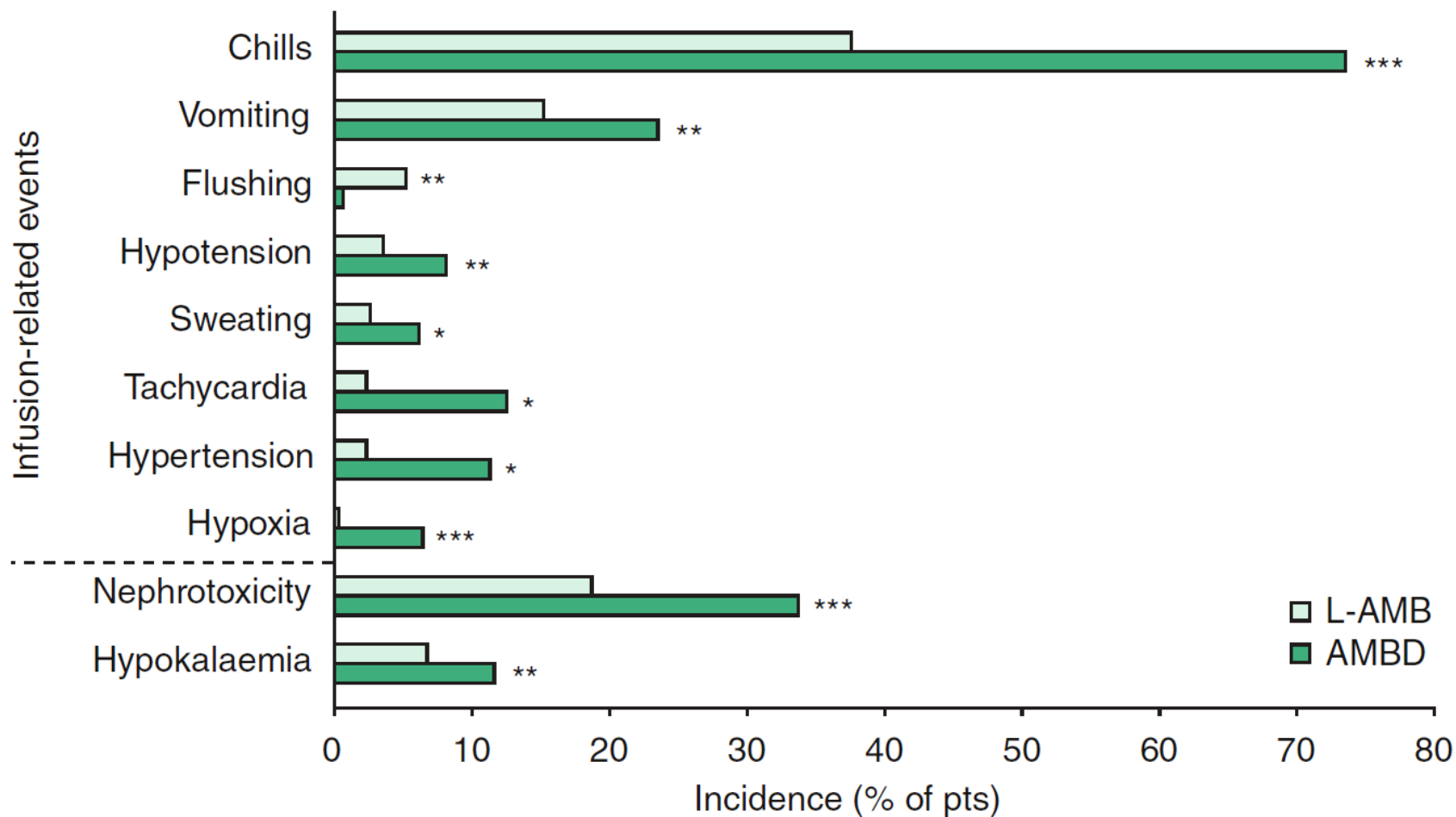
Liposomal Amphotericin B

Organ	Tissue concentration (Compared with Conventional AmphoB)
Serum	> 5-20 folds (data from peak serum Conc.)
Liver	0.5-1 fold
Spleen	3 folds (esp. in reticuloendothelial system) • May promote drug delivery to infected site
Lung	0.2 fold
Kidney	0.2 fold • Plays a part to reduce nephrotoxicity
Brain	1.2 folds





Safety Profiles of Liposomal Amphotericin B



L-AMB = Liposomal amphotericin B

AMBD = Amphotericin B deoxycholate (Conventional)



Amphotericin B: Drug Interactions

- **Nephrotoxic agents**
 - Immunosuppressants: Cyclosporine, Tacrolimus
 - Antibacterials: Aminoglycosides, Vancomycin, Teicoplanin
 - Anticancers: Cisplatin, Carboplatin, Oxaliplatin
 - Others: Diuretics, Iodinated contrast media, etc.
- **Neuromuscular-blocking agents:**
 - Atracurium, Vecuronium, etc.
 - AmphoB-induced HypoK → ↑ risk of respiratory depression
- **Digoxin** (narrow therapeutic window)
 - AmphoB-induced HypoK → ↑ risk of digoxin toxicity



Nystatin

- Poorly absorbed through GI, skin, mucosa
- Also toxic if parenterally administered



- Topical uses for non-invasive candidiasis
 - Oropharynx
 - Esophagus
 - Vagina
- Oral suspension
(swish and swallow)
- Vaginal suppository

Natamycin (Pimaricin)

- Poorly absorbed through GI



Ophthalmic suspension (Eye drop)

- Effective conc. at corneal stroma
- But not intraocular fluid



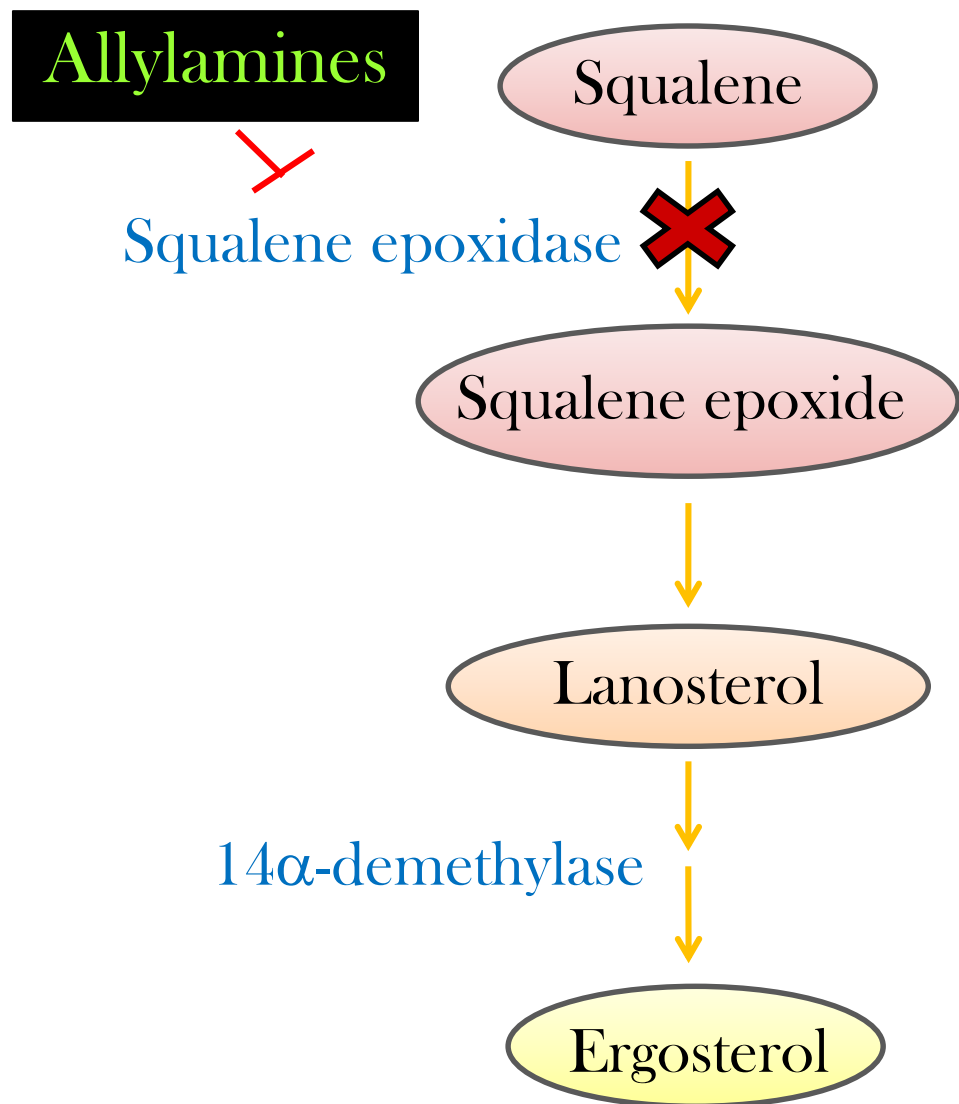
Blepharitis, Conjunctivitis, Keratitis

- Fungicidal, Conc.-dependent activity
 - Against *Candida*, *Aspergillus*, *Fusarium*



Allylamines

- **Drug:** Terbinafine
Naftifine
 - **Mechanism of action**
 - Inhibit squalene epoxidase enzyme
- Inhibit ergosterol synthesis** **Toxic squalene accumulation**
- Fungal cell death**





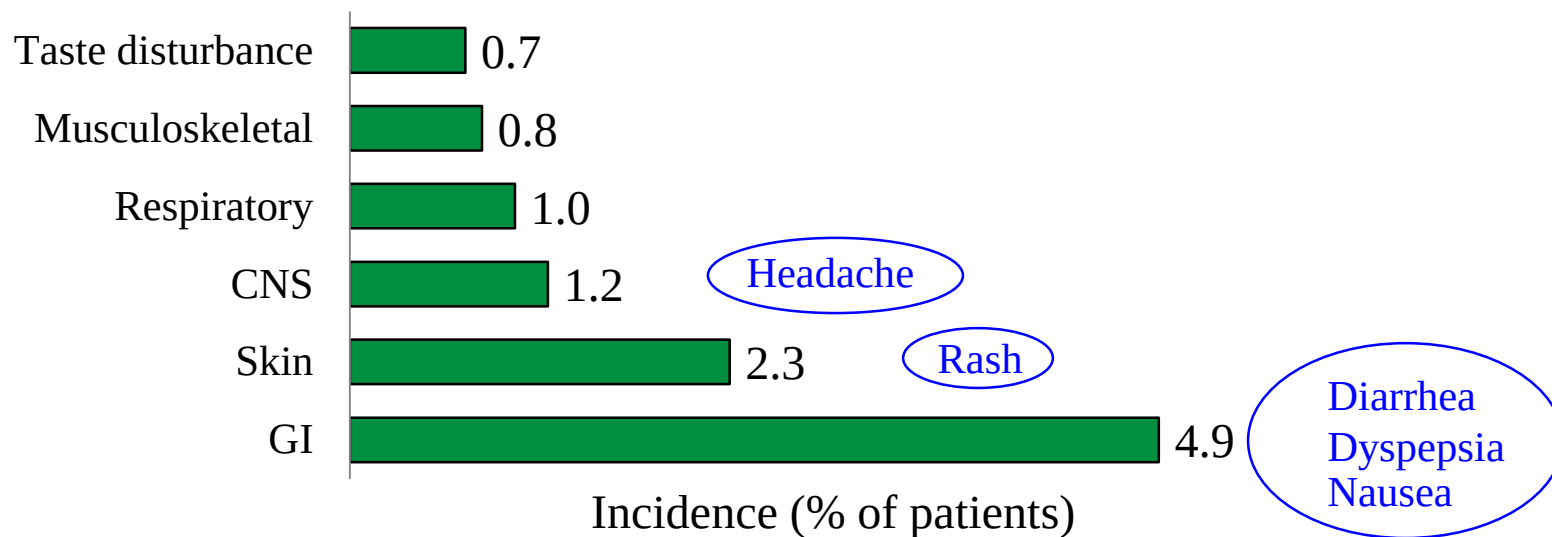
Terbinafine

- Oral bioavailability ~ 40% (first-pass met.)
- Fungicidal for dermatophytes, Fungistatic for *C. albicans*
- Accumulates in (and slow release from) skin, nail, fat
 - Drug of choice for
 - Dermatophytoses, esp., **onychomycosis** (nail infection)
 - Subcutaneous mycoses
- Low blood concentration: No role for systemic mycoses
- $t_{1/2}$: Initial ~12 hrs, At steady state: 200-400 hrs → OD dosing
 - Not recommended for severe hepatic and renal insufficiency



Terbinafine (Cont.)

- ADRs



McClellan KJ, et al. Drugs 1999;58(1):179-202.

- **Drug interactions:** Terbinafine is metabolized by at least 7 CYPs
 - Rifampicin (enz. inducer) reduces terbinafine level
 - Cimetidine (enz. inhibitor) increases terbinafine level

Azoles

- Mechanism of action

- Inhibit 14α -demethylase enzyme (fungal CYP)

Inhibit ergosterol synthesis

Toxic lanosterol accumulation

Fungal cell death

Allylamines

Squalene epoxidase

Squalene

Squalene epoxide

Lanosterol

Azoles

14α -demethylase

Ergosterol





Azoles (Cont.)

Imidazoles	Triazoles
• Ketoconazole • Clotrimazole • Isoconazole • Miconazole • Econazole • Sertaconazole • Tioconazole • Luliconazole Topical uses	• 1st Gen -Fluconazole (Diflucan[®]) -Itraconazole (Sporal[®]) • 2nd Gen (extended-spectrum) -Voriconazole (Vfend[®]) -Posaconazole (Noxafil[®]) -Isavuconazole (Cresemba[®]) Systemic antifungals

- Triazoles have less effect on human sterol synthesis and CYPs



“Less likely to be toxic” for systemic uses



Ketoconazole

- Orally active (tablet dosage form is available)
 - Lipophilic weak base
 - Dissolution and absorption require acidic pH

Henderson-Hasselbalch equation;

$$\log \frac{(\text{Protonated})}{(\text{Unprotonated})} = \text{pK}_a - \text{pH}$$



$$\log \frac{(\text{Protonated})}{(\text{Unprotonated})} = 6.5 - 2.5 = 4$$

$$\frac{(\text{Protonated})}{(\text{Unprotonated})} = 10^4$$

Dissolved



Ketoconazole (Cont.1)

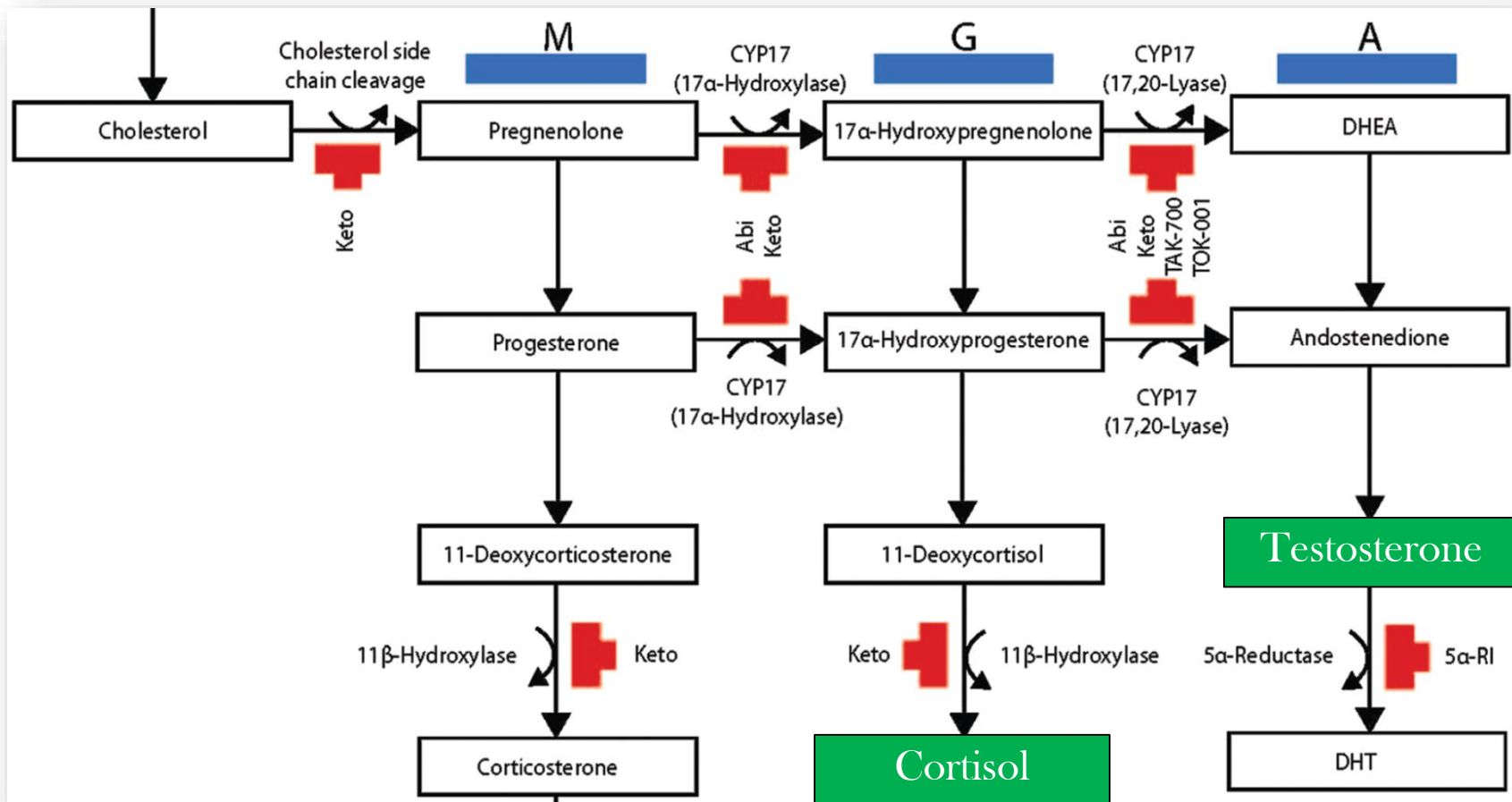
- To achieve oral bioavailability
 - Taken with food
 - Stimulate gastric acid secretion
 - Slow gastric emptying rate
 - Avoid drugs that increase gastric pH
 - Antacids
 - H₂-blockers
 - Proton-pump inhibitors
 - Cola drinks (pH~2.5) improve drug absorption in patients with achlorhydria (no or low HCl secretion)

Ketoconazole (Cont.2)

BUT

- Oral formulations are no longer used as first-line drug
 - Potentially fatal liver injury
 - Adrenal insufficiency
 - ↓ Corticosteroids & Testosterone
 - Harmful drug interactions
 - Strong CYP3A4, CYP2C9, CYP1A2 inhibitor

Ketoconazole (Cont.3)



Schweizer MT and Antonarakis ES. Ther Adv Urol 2012;4(4):167-78.

- High dose of oral ketoconazole is used to treat Cushing disease and sometimes Prostate cancer

Fluconazole

- Available as both IV and oral forms
- Hydrophilic weak base (highly water soluble)
- Good oral bioavailability ($> 90\%$)
 - Not affected by gastric pH and food
 - Allow switching between IV and oral forms
- Predictable, linear PK





Fluconazole (Cont.)

- Widely distributed with high conc. in
 - CSF ($\approx 80\%$ of serum level) $\rightarrow$ Brain infections
 - Vitreous humor (28-75% of serum level) $\rightarrow$ Eye inf.
- Long $t_{1/2}$ (≈ 30 hrs) $\rightarrow$ OD dosing
- Excreted via renal with 70-80 % unchanged form
 $\rightarrow$ Urinary tract infections

Itraconazole

- Active metabolite: **Hydroxy-itraconazole**
 - Equivalent potency with parent drug
- Highly lipophilic weak base ($pK_a = 3.7$)



Capsule formulation:

- Requires acidic pH and /or food for optimal dissolution

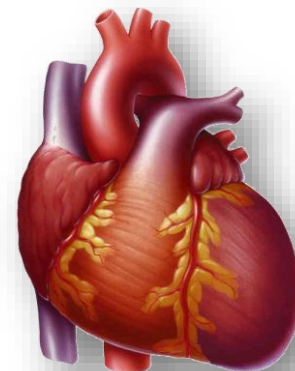
Oral Solution: Cyclodextrin as a solubilizing agent

- Administered on an empty stomach
- Oral bioavailability: **first-pass met.**
 - $\approx 22\%$ (capsule) and 55% (solution)



Itraconazole (Cont.)

- Widely distributed
 - Adipose tissue → Cutaneous and Subcutaneous infections
 - Bone, Except CSF
- IV itraconazole
 - In a healthy human: transient asymptomatic decreases in left ventricular ejection fraction
 - May result in manifestation or worsening of signs and symptoms of congestive heart failure (CHF)
- Oral formulations
 - Also not recommended in patients with history of ventricular dysfunction e.g. CHF



Voriconazole

- Hydrophilic weak base (highly water soluble)
 - Derivative of Fluconazole
- Good oral bioavailability (> 90%)
 - Not affected by gastric pH
 - Fatty diets slightly ↓ bioavailability to $\approx 80\%$
 - Allow switching between IV and oral forms
- Widely distributed with high conc. in
 - CSF ($\approx 60\%$ of serum level) and higher in brain tissue
 - Vitreous humor (38% of serum level)



Voriconazole (Cont.)

- In adults, non-linear PK (saturation of metabolism; CYP2C19)
- In children, linear PK with more rapid metabolism
 - Higher dose of drug is needed
- CYP2C19 polymorphisms $\approx$ 13% of Thai is poor met.)
- ADRs
 - Visual disturbances (30%)
 - Occur during the first week $\rightarrow$ disappear when continued therapy
 - Altered color discrimination, blurred vision, appearance of bright spots and wavy lines, photophobia
 - Mild rash (17%), including photosensitivity
 - $\uparrow$ Liver enzymes (10%), like other triazoles





Posaconazole

- Lipophilic weak base ($\text{pK}_{\text{a}1} = 3.6$, $\text{pK}_{\text{a}2} = 4.6$)
 - Similar structure to Itraconazole
- Intravenous
- Gastro-resistant tablet (pH-dependent polymer designed to inhibit the release of posaconazole until it reaches the small intestine) → Avoid low bioavailability
- Oral suspension (low bioavailability: 8-47%)
 - Requires acidic pH and /or fatty food
 - Absorption is saturated → BID-QID dosing with meal
- Results from clinical studies and case reports indicated it is widely distributed
 - CNS, Bone, Eye
- 20-30% is metabolized by UGT1A4
- 77% is excreted via feces (66% as unchanged)



Isavuconazole

- Available as prodrug (IV and oral formulations)
 - Isavuconazonium (sulfate) $\xrightarrow{\text{esterases}}$ Isavuconazole
- High oral bioavailability (>95%)
 - No pH & food effects
- Widely distributed, including vitreous humor
 - Low level in CSF, but high level in brain
- Long half-life (56-104 hrs) → Once daily
- Urinary excretion is very low
- Approved in 2015 for invasive aspergillosis and invasive mucormycosis



Triazoles: Spectrum of Activity

Organism	Triazole Antifungals				
	Flu	Itr	Vor	Pos	Isa
<i>Candida</i> spp.					
<i>C. albicans</i>	+	+	+	+	+
<i>C. glabrata</i> (efflux pump)	±	±	±	±	±
<i>C. krusei</i> (impaired binding with 14 α -demethylase)	-	±	+	+	+
<i>C. lusitaniae</i>	+	+	+	+	+
<i>C. parapsilosis</i>	+	+	+	+	+
<i>C. tropicalis</i>	+	+	+	+	+
<i>C. guilliermondii</i>	±	±	+	+	+

Note: + has activity, - does not have activity, ± variable activity



Triazoles: Spectrum of Activity (Cont.1)

Organism	Triazole Antifungals				
	Flu	Itr	Vor	Pos	Isa
<i>Cryptococcus neoformans</i>	+	+	+	+	+
<i>Aspergillus</i> spp.					
<i>A. flavus</i>	-	+	+	+	+
<i>A. fumigatus</i>	-	+	+	+	+
<i>A. niger</i>	-	±	+	+	+
<i>A. terreus</i>	-	+	+	+	+
<i>A. nidulan</i>	-	+	+	+	+

Note: + has activity, - does not have activity, ± variable activity



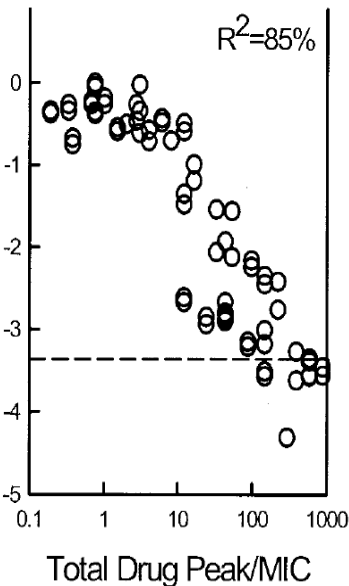
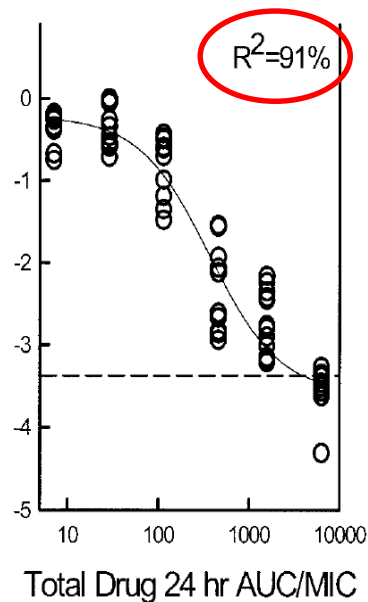
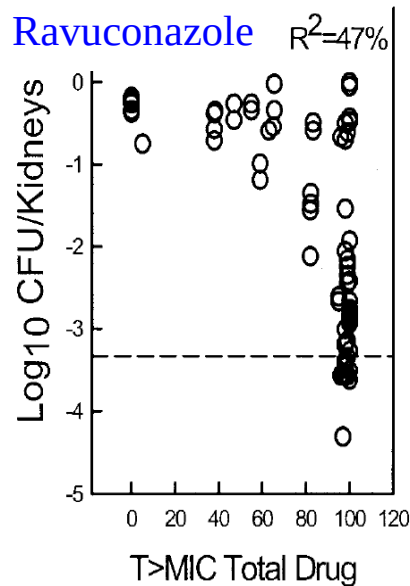
Triazoles: Spectrum of Activity (Cont.2)

Organism	Triazole Antifungals				
	Flu	Itr	Vor	Pos	Isa
<i>Coccidioides</i> spp.	+	+	+	+	+
<i>Blastomyces</i> (genus)	+	+	+	+	+
<i>Histoplasma</i> spp.	+	+	+	+	+
<i>Fusarium</i> spp.	-	-	+	+	+
<i>Scedosporium apiospermum</i> Or <i>Pseudallescheria boydii</i>	-	±	+	+	+
Zygomycetes (class), including Mucorales (order)	-	-	-	+	+

Note: + has activity, - does not have activity, ± variable activity

Triazoles: Pattern of Activity

- Generally fungistatic
 - But Fungicidal activity against *Aspergillus* spp.
 - Except Fluconazole (no activity)
- Time-dependent killing activity with prolonged PAE



Flu, Itr, Vor:

- AUC:MIC > 25

Pos:

- AUC:MIC > 400

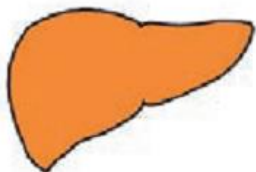
Isa:

- AUC:MIC > 270



Triazoles: Summary of ADRs

Hepatic



All azoles

Renal toxicity



Cyclodextrins possibly
toxic (IV voriconazole)

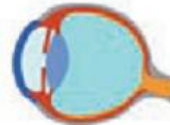
CNS



Voriconazole

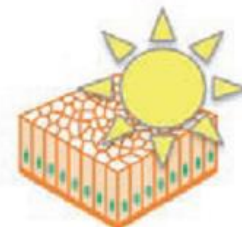
- Headache
- Confusion
- Hallucination

Photopsia



Voriconazole

Cutaneous



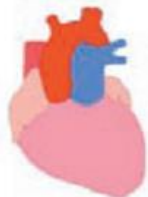
Rash (all antifungal agents)
Photosensitivity/malignancy?
(voriconazole)

GI



Itraconazole
Posaconazole

Cardiac



Cardiomyopathy
(itraconazole)

QTc prolongation
(all azoles, especially
with drug interactions)



Pharmacokinetic Drug Interactions of Triazoles

Mechanism	Flu- conazole	Itra- conazole	Posa- conazole	Vori- conazole
Inhibitor				
CYP2C19	+	—	—	+++
CYP2C9	++	+	—	++
CYP3A4	++	+++	+++	++
Substrate				
CYP2C19	—	—	—	+++
CYP2C9	—	—	—	+
CYP3A4	+	+++	—	+

— = no activity; + = minimal activity; ++ = moderate activity; +++ = strong activity.

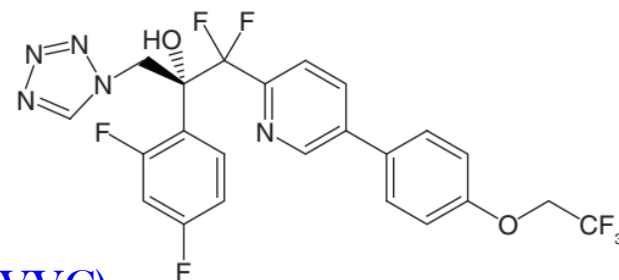
Lewis RE. Mayo Clin Proc 2011; 86(8):805-17.

- Isavuconazole is a substrate of CYP3A4
- Itraconazole and Posaconazole also inhibit P-gp



Oteseconazole

- Orally active **tetrazole** (BA ~40%)
- High-fat, high-calorie meal increases absorption of capsules
- Not metabolized and No CYP interaction
- Half-life ~138 days
- Approved in 2021 for **recurrent vulvovaginal candidiasis (RVVC)**



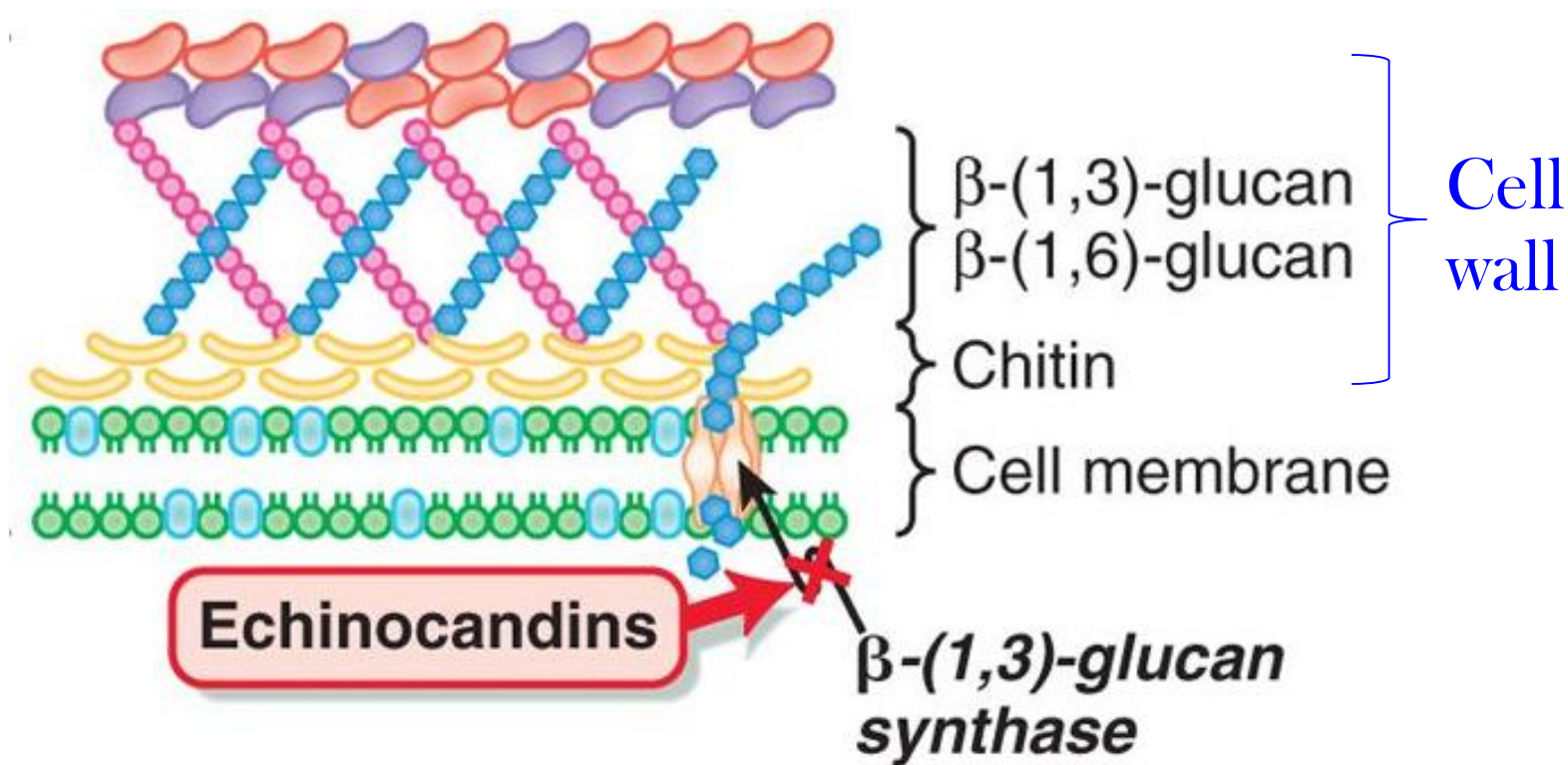
Candida clinical isolate [type] (no. tested)	MIC ₅₀ or MIC range (μg/ml) at 24 h [†]	
	Oteseconazole	Fluconazole
<i>C. albicans</i> [WT] (10)	≤0.03	≤0.125
<i>C. albicans</i> [azole resistant] (10)	1	>64
<i>C. glabrata</i> [WT] (10)	0.06	1
<i>C. glabrata</i> [azole resistant] (10)	2	>64
<i>C. parapsilosis</i> (10)	≤0.03	0.125
<i>C. guilliermondii</i> (3)	0.06–0.125	1–4
<i>C. krusei</i> (3)	0.125–0.25	16–32
<i>C. lusitaniae</i> (3)	≤0.03–0.25	≤0.125–0.25
<i>C. tropicalis</i> (5)	≤0.03–4	≤0.125–1
Isolate from acute VVC (no. tested)	MIC range (μg/ml)	
	Oteseconazole	Fluconazole
<i>C. albicans</i> (41)	≤0.001–0.06	≤0.12–4
<i>C. parapsilosis</i> (12)	≤0.001–0.12	0.25–2
<i>C. tropicalis</i> , <i>C. lusitaniae</i> , <i>C. glabrata</i> (7)	≤0.002–1	0.25–64

[†]Data values for which ranges are not provided indicate that the isolates responded to the same dilution.

MIC: Minimum inhibitory concentration; MIC₅₀: Minimum concentration that caused inhibition of 50% of the isolates tested; VVC: Vulvovaginal candidiasis; WT: Wild-type.

Echinocandins

- **Drugs:** Caspofungin, Micafungin, Anidulafungin, Rezafungin
- Mechanism of action: Disrupt cell wall rigidity





Echinocandins: Comparison of Pharmacological Properties

Properties	Caspofungin	Micafungin	Anidulafungin	Rezafungin
Oral bioavailability (%)	< 10	< 10	2-7	<10
Protein binding (%)	96-97	99.8	>99	97.4
Half-life (h)	9-11	11-17	24-26	80 (first dose) 150 (2 nd &3 rd dose) Once weekly use
CSF penetration	Low	Low	< 0.1% plasma	Low
Metabolism	Slow peptide hydrolysis, N-acetylation and spontaneous degradation to inactive product	Catechol-O-methyltransferase pathway	• Not metabolized • Slow chemical degradation to inactive metabolites	• Metabolized by hydroxylation of the terphenyl, pentyl ether side chain



Echinocandins: Spectrum of Activity

Organism	Echinocandins			
	Cas	Mic	Ani	Rez
<i>Candida</i> spp.				
<i>C. albicans</i>	+	+	+	+
<i>C. glabrata</i>	+	+	+	+
<i>C. krusei</i>	+	+	+	+
<i>C. lusitaniae</i>	+	+	+	+
<i>C. parapsilosis</i>	±	±	±	±
<i>C. tropicalis</i>	+	+	+	+
<i>C. guilliermondii</i>	±	±	±	±

Note: + has activity, - does not have activity, ± variable activity



Echinocandins: Spectrum of Activity (Cont.1)

Organism	Echinocandins			
	Cas	Mic	Ani	Rez
<i>Cryptococcus neoformans</i>	-	-	-	-
<i>Aspergillus</i> spp.				
<i>A. flavus</i>	+	+	+	+
<i>A. fumigatus</i>	+	+	+	+
<i>A. niger</i>	+	+	+	+
<i>A. terreus</i>	+	+	+	+
<i>A. nidulan</i>	+	+	+	+

Note: + has activity, - does not have activity, ± variable activity



Echinocandins: Spectrum of Activity (Cont.2)

Organism	Echinocandins			
	Cas	Mic	Ani	Rez
<i>Coccidioides</i> spp.	±	±	±	±
<i>Blastomyces</i> (genus)	±	±	±	±
<i>Histoplasma</i> spp.	±	±	±	±
<i>Fusarium</i> spp.	-	-	-	-
<i>Scedosporium apiospermum</i> Or <i>Pseudallescheria boydii</i>	-	-	-	-
Zygomycetes (class), including Mucorales (order)	-	-	-	-

Note: + has activity, - does not have activity, ± variable activity

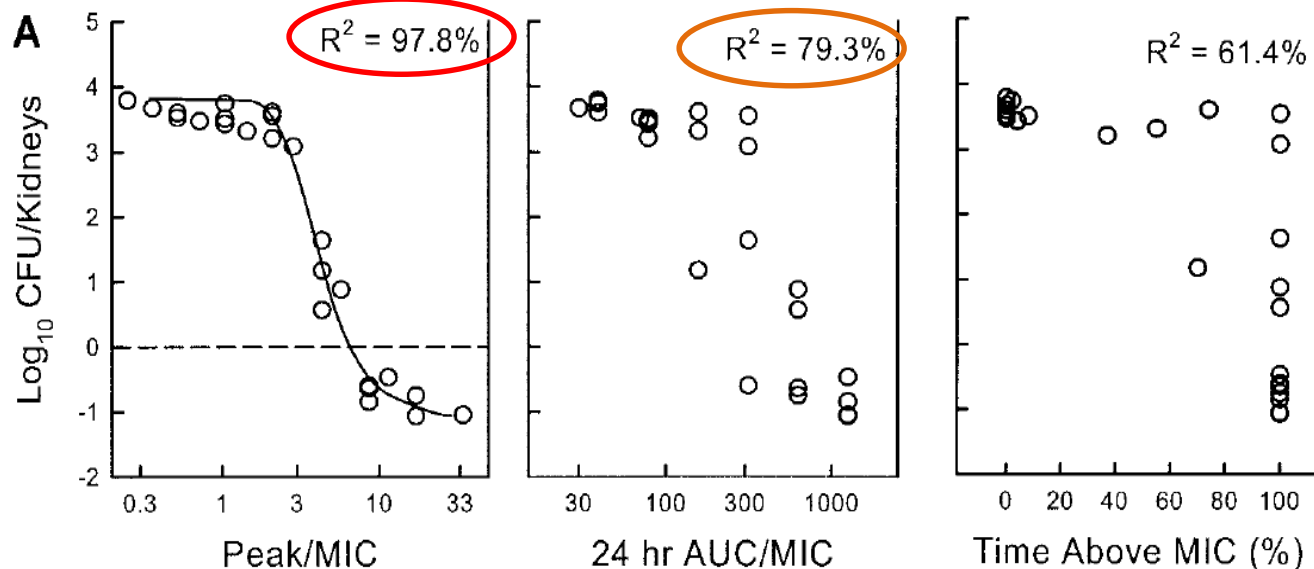


Echinocandins: Pattern of Activity

- Fungicidal for *Candida* spp.

But Fungistatic for *Aspergillus* spp.

- Conc-dependent killing activity with prolonged PAE



- $C_{max}:MIC > 10$
serum (unbound)

Or

- $AUC:MIC > 20$

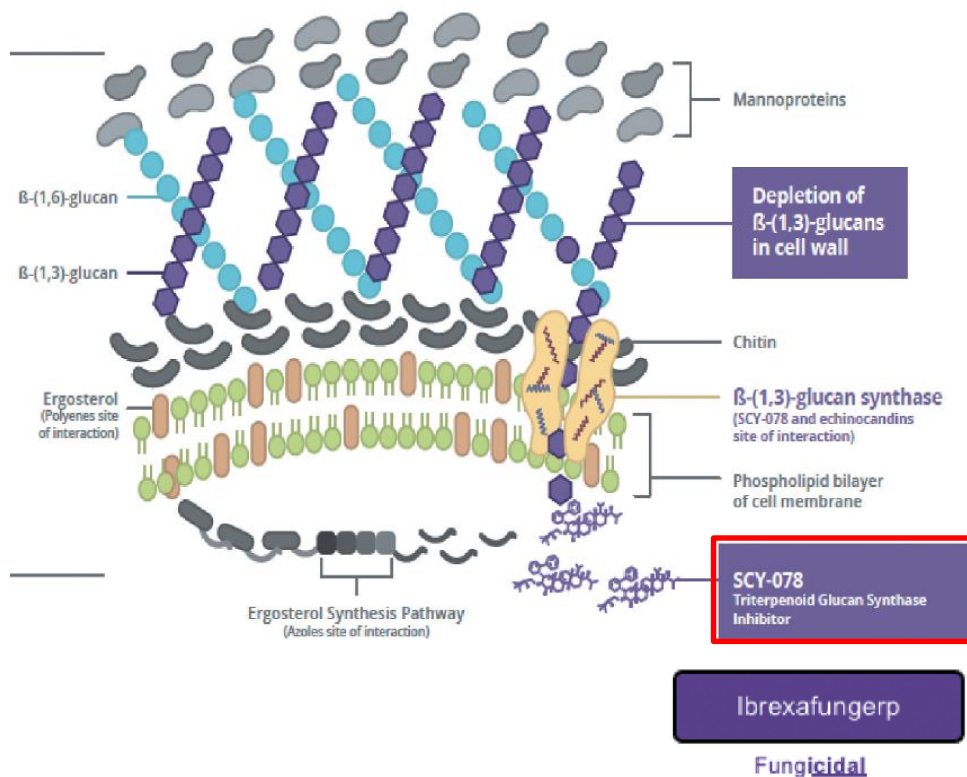


Echinocandins:

ADRs and Drug Interactions

- Less toxic (selective toxicity to fungal cell wall)
 - Infusion-related reactions:
 - **Phlebitis** (vein inflammation)
 - **Histamine-like effects** in cases of rapid infusion
(flushing, urticaria, bronchospasm, facial swelling, pruritis)
- **Drug interactions**
 - Caspofungin: ↓ Tacrolimus level (~20%)
 - Micafungin: mild CYP3A4 inhibitor
 - ↑ AUC of Nifedipine (18%) and Sirolimus (21%)

Ibrexafungerb



- Inhibits glucan synthase
- Orally active, metabolized by CYP3A4
- Half-life of approximately 20 to 30 hours
- For the treatment of vulvovaginal candidiasis (VVC) and prevention of recurrence of VVC
- Most common ADRs:
 - GI (abdominal discomfort, diarrhea, nausea, and vomiting)
 - Headache

Efficacy	Active vs. <i>Candida albicans</i>	●	●
	Active vs. non- <i>albicans Candida</i>	●	●
	Active vs. azole-resistant	●	○
	Enhanced activity at acidic pH	●	○
	Distribution into vaginal tissue	●	●



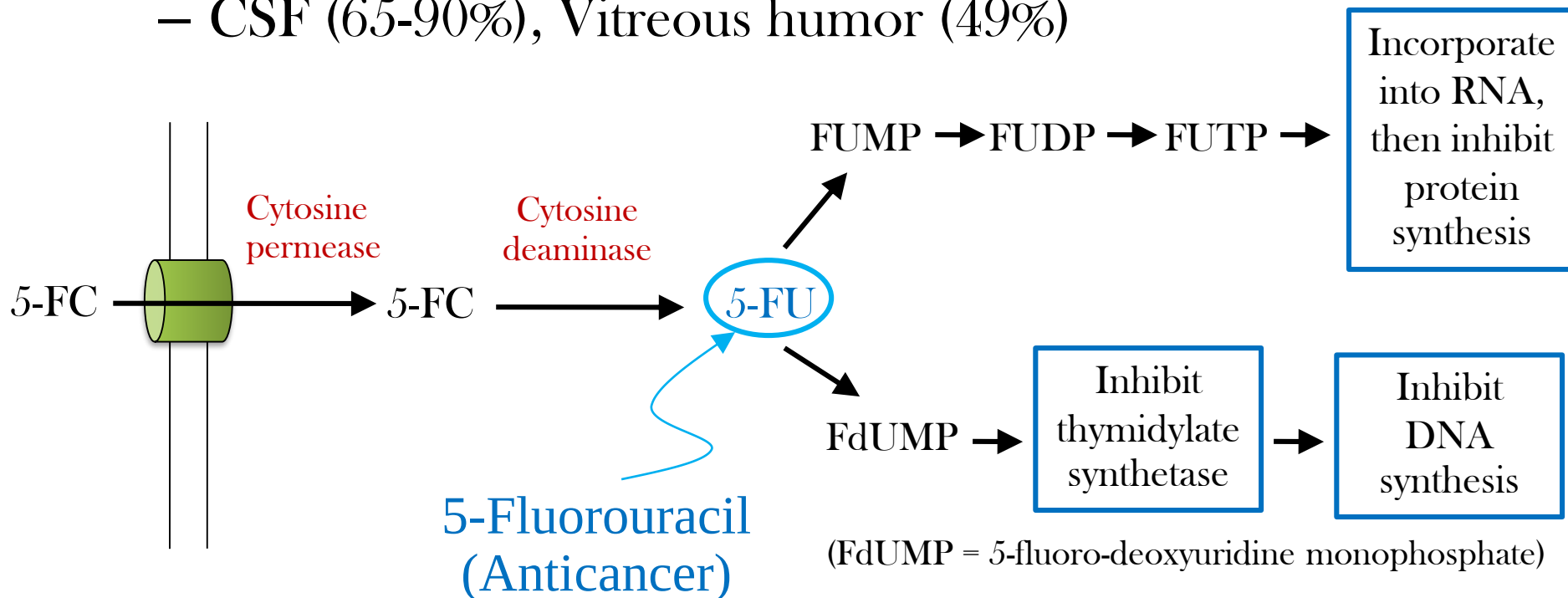
Effects of vaginal pH on ibrexafungerp in vitro activity

Antifungal	pH 7.0	pH 5.72	pH 4.5
<i>C. galbrata</i> MIC ₅₀ µg/mL			
Ibrexafungerp	0.5–1	0.5	0.031–0.063
Fluconazole	0.05–2	2–16	1–16
<i>C. albicans</i> MIC ₅₀ µg/mL			
Ibrexafungerp	0.125–0.5	0.125–0.25	0.16–0.031
Fluconazole	<0.125–1	<0.125–1	0.25–8

MIC = minimum inhibitor concentration

Flucytosine

- Flucytosine = 5-fluorocytosine (5-FC)
 - Pyrimidine analogue
- Well absorbed from GI tract and widely distributed
 - CSF (65-90%), Vitreous humor (49%)





Flucytosine (Cont.)

- ~ 80% is excreted unchanged via renal route
- Fungistatic, Time-dependent killing activity without PAE
 - T > MIC 20-40%
 - Against only *Candida* spp. and *Cryptococcus* spp.
- High rate of resistance → Use in combinations
- ADRs
 - GI microflora possibly converts 5-FC to 5-FU
 - Nausea, vomiting, enterocolitis
 - Bone marrow suppression: Leukopenia, Thrombocytopenia



Griseofulvin

- Mechanism of action: inhibit microtubule function → disrupt the assembly of the mitotic spindle → inhibit fungal mitosis
- Fungistatic against dermatophytes
- Orally active (high-fat diets ↑ drug absorption)
- Accumulates in skin, hair, and nail
 - Used in Dermatophytoses
 - No role in systemic mycoses
- Induces CYP3A4 → e.g. ↓ Warfarin level
- Disulfiram-like reaction when taking with alcohol



Examples of Topical Antifungals

Drug Class	Drug Name	Preparation
Azoles	Clotrimazole	Cream and Vaginal tab. (Canesten [®]) Troche
	Econazole (Econ [®])	Cream
	Isoconazole (Travogen [®])	Cream
	Ketoconazole	Cream and Shampoo (Nizoral [®]) Lotion
	Miconazole (Daktarin [®])	Cream, Oral gel
	Sertaconazole (Zalain [®])	Cream, Vaginal tab.
	Tioconazole (Trosyd [®])	Cream
Allylamines	Terbinafine (Lamisil [®])	Cream
Polyenes	Nystatin	Vaginal tab. (Gynecon [®]) Oral susp. (Mycostatin [®]) Combined Cream (Dermacombin [®])



Examples of Topical Antifungals (Cont.)

Drug Class	Drug Name	Preparation
Others	Tolnaftate (Tonaf [®])	Cream
	Amorolfine (Loceryl [®])	Nail Lacquer
	Ciclopirox (Loprox [®])	Solution
	Tavaborole (Kerydin [®])	Solution
	Salicylic acid (keratolytic) ± Benzoic acid (fungistatic)	Lotion (Zema [®] , Skindure [®]) Ointment (Whitfield's ointment, ขี้ผึ้งเปออร์ 28 [®])
	Undecylenic acid	Cream, Ointment
	Gentian violet	Solution
	Sodium thiosulphate	Crystal + H ₂ O → Solution
	Selenium sulfide (Selsun [®])	Shampoo

Thank You!

